

Newman-Penrose Formalism Calculations

Stephani et al., Equation (28.56c)

Metric

$$ds^2 = \left(-\frac{4a\kappa_0 r x + 2a\kappa_0 u e^x + a\kappa_0 r \log(|au + b|) + 2b\kappa_0 e^x - 8m_0 r}{4r^2} \right) du \otimes du - du \otimes dr - dr \otimes du + r^2 e^{(-x)} dx \otimes dx + r^2 e^{(-x)} dy \otimes dy$$

Coordinates

$$(x^\mu) = (u, r, x, y)$$

Null Tetrad

Covariant components

$$l = (1) du$$

$$n = \left(\frac{a\kappa_0 x}{2r} + \frac{a\kappa_0 u e^x}{4r^2} + \frac{a\kappa_0 \log(|au + b|)}{8r} + \frac{b\kappa_0 e^x}{4r^2} - \frac{m_0}{r} \right) du + (1) dr$$

$$m = \left(\frac{1}{2} \sqrt{2} r e^{(-\frac{1}{2}x)} \right) dx + \left(-\frac{1}{2} i \sqrt{2} r e^{(-\frac{1}{2}x)} \right) dy$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} r e^{(-\frac{1}{2}x)} \right) dx + \left(\frac{1}{2} i \sqrt{2} r e^{(-\frac{1}{2}x)} \right) dy$$

Contravariant components

$$l = (-1) dr$$

$$n = (-1) du + \left(\frac{a\kappa_0 x}{2r} + \frac{a\kappa_0 u e^x}{4r^2} + \frac{a\kappa_0 \log(|au + b|)}{8r} + \frac{b\kappa_0 e^x}{4r^2} - \frac{m_0}{r} \right) dr$$

$$m = \left(\frac{\sqrt{2}e^{\left(\frac{1}{2}x\right)}}{2r} \right) dx + \left(-\frac{i\sqrt{2}e^{\left(\frac{1}{2}x\right)}}{2r} \right) dy$$

$$\overline{m} = \left(\frac{\sqrt{2}e^{\left(\frac{1}{2}x\right)}}{2r} \right) dx + \left(\frac{i\sqrt{2}e^{\left(\frac{1}{2}x\right)}}{2r} \right) dy$$

Directional Derivatives

$$D = -\partial_r$$

$$\Delta = -\partial_u + \frac{a\kappa_0 x}{2r} + \frac{a\kappa_0 u e^x}{4r^2} + \frac{a\kappa_0 \log(|au+b|)}{8r} + \frac{b\kappa_0 e^x}{4r^2} - \frac{m_0}{r} \partial_r$$

$$\delta = \frac{\sqrt{2}e^{\left(\frac{1}{2}x\right)}}{2r} \partial_x - \frac{i\sqrt{2}e^{\left(\frac{1}{2}x\right)}}{2r} \partial_y$$

$$\overline{\delta} = \frac{\sqrt{2}e^{\left(\frac{1}{2}x\right)}}{2r} \partial_x + \frac{i\sqrt{2}e^{\left(\frac{1}{2}x\right)}}{2r} \partial_y$$

Spin Coefficients

$$\rho = \frac{1}{r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{a\kappa_0 x}{2r^2} + \frac{a\kappa_0 u e^x}{4r^3} + \frac{a\kappa_0 \log(|au+b|)}{8r^2} + \frac{b\kappa_0 e^x}{4r^3} - \frac{m_0}{r^2}$$

$$\lambda = 0$$

$$\nu = \frac{\sqrt{2}a\kappa_0 u e^{\left(\frac{3}{2}x\right)}}{8r^3} + \frac{\sqrt{2}a\kappa_0 e^{\left(\frac{1}{2}x\right)}}{4r^2} + \frac{\sqrt{2}b\kappa_0 e^{\left(\frac{3}{2}x\right)}}{8r^3}$$

$$\alpha = \frac{\sqrt{2}e^{\left(\frac{1}{2}x\right)}}{8r}$$

$$\beta = -\frac{\sqrt{2}e^{\left(\frac{1}{2}x\right)}}{8r}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = \frac{a\kappa_0 x}{4r^2} + \frac{a\kappa_0 u e^x}{4r^3} + \frac{a\kappa_0 \log(|au + b|)}{16r^2} + \frac{b\kappa_0 e^x}{4r^3} - \frac{m_0}{2r^2}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{a\kappa_0 u e^x}{4r^4} + \frac{b\kappa_0 e^x}{4r^4}$$

$$\Phi_{12} = \frac{\sqrt{2}a\kappa_0 u e^{(\frac{3}{2}x)}}{8r^4} + \frac{\sqrt{2}a\kappa_0 e^{(\frac{1}{2}x)}}{8r^3} + \frac{\sqrt{2}b\kappa_0 e^{(\frac{3}{2}x)}}{8r^4}$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = \frac{\sqrt{2}a\kappa_0 u e^{(\frac{3}{2}x)}}{8r^4} + \frac{\sqrt{2}a\kappa_0 e^{(\frac{1}{2}x)}}{8r^3} + \frac{\sqrt{2}b\kappa_0 e^{(\frac{3}{2}x)}}{8r^4}$$

$$\Phi_{22} = \frac{a^2\kappa_0 u^2 e^{(2x)}}{8(ar^4u + br^4)} + \frac{a^2\kappa_0 r u e^x}{4(ar^4u + br^4)} + \frac{a^2\kappa_0 r^2}{8(ar^4u + br^4)} + \frac{ab\kappa_0 u e^{(2x)}}{4(ar^4u + br^4)} + \frac{ab\kappa_0 r e^x}{4(ar^4u + br^4)} + \frac{b^2\kappa_0 e^{(2x)}}{8(ar^4u + br^4)}$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{a\kappa_0 x}{2r^3} + \frac{a\kappa_0 u e^x}{2r^4} + \frac{a\kappa_0 \log(|au + b|)}{8r^3} + \frac{b\kappa_0 e^x}{2r^4} - \frac{m_0}{r^3}$$

$$\Psi_3 = \frac{\sqrt{2}a\kappa_0 u e^{(\frac{3}{2}x)}}{4r^4} + \frac{3\sqrt{2}a\kappa_0 e^{(\frac{1}{2}x)}}{8r^3} + \frac{\sqrt{2}b\kappa_0 e^{(\frac{3}{2}x)}}{4r^4}$$

$$\Psi_4 = \frac{a\kappa_0 u e^{(2x)}}{4r^4} + \frac{a\kappa_0 e^x}{4r^3} + \frac{b\kappa_0 e^{(2x)}}{4r^4}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"